

AHAMED RAAFIQ

ELECTRONICS ENGINEER — ROBOTICS & AUTONOMOUS SYSTEMS

Colombo, Sri Lanka | raafiqofficial@gmail.com | +94 766994358

[linkedin.com/in/ahamedraafiq](https://www.linkedin.com/in/ahamedraafiq) | github.com/AhamedRaafiq

PROFESSIONAL SUMMARY

Electronics Engineering graduate passionate about Robotics & Autonomous Systems, with hands-on experience spanning robotics, embedded and control systems, electronics hardware and PCB design, IoT, sensor fusion, digital signal processing, biomedical instrumentation, and UAVs. Driven to grow within research and development roles rather than routine engineering positions, with a strong interest in continuously learning and going deeper into these fields through applied, hands-on work.

EDUCATION

BSc in Electronics Engineering

2022 – 2026

University of Engineering and Technology, Taxila

G.C.E. Advanced Level

2021

Royal College, Colombo, Sri Lanka

SKILLS

Robotics & Automation: ROS2 (Python, C++), Gazebo simulation, URDF/transform trees, TF2, Sensor Fusion (Madgwick AHRS), Nav2 path planning, multi-DOF robotic arm control & motion planning

Control Systems & Instrumentation: PID tuning (MATLAB), feedback control systems, sensor & actuator integration, real-time data acquisition, instrumentation & measurement

Electronics & PCB Design: Altium Designer, Cadence, Proteus, Multisim, schematic capture & PCB layout, component selection, power integrity, Gerber/BOM generation

Embedded Systems & Hardware Integration: Embedded C/C++, microcontroller firmware (Arduino, ESP32-S3), Raspberry Pi, PWM motor control, IMU Sensor (MPU6050), Battery Management System with 18650cells, H-bridge & motor drivers (TB6612FNG, BCA8016), buck converters (XL4016), servo control, I2C/UART/USB-CDC/SPI, ESP-IDF

UAV & Drone Systems: PX4 flight-controller firmware, QGroundControl mission & manual control, flight parameter tuning, UAV avionics, telemetry capture

Machine Learning: XGBoost regression, feature engineering & scaling, hyperparameter tuning, model evaluation (MAE, RMSE, R^2 , MAPE, Clarke Grid), data validation & normalization

Digital Signal Processing: Time/frequency-domain filtering, noise reduction, ensemble averaging, sensor signal acquisition & feature extraction, Python-based DSP pipelines (NumPy, SciPy), Biomedical Signal Processing (PPG)

Hardware Testing & Commissioning: Oscilloscope, multimeter, soldering, bench power supply, hardware bring-up, testing, debugging & troubleshooting, prototype validation, technical documentation

Programming & Tools: C, C++ (embedded firmware, OOP, STL), Python (NumPy, SciPy, Pandas), MATLAB, Git/GitHub, VS Code, Ubuntu Linux

EXPERIENCE

Digital Signal Processing Intern

Jun 2025 – Aug 2025

University of Engineering and Technology, Taxila. (On-site)

- Built Python-based DSP pipelines for audio signal processing, filtering, transforms, and noise reduction.
- Performed signal feature extraction and testing to support downstream analysis, classification tasks, and technical reporting.
- Applied Python (NumPy, SciPy) for real-time signal handling, data processing, and visualisation.

Robotics & Simulation Intern

Apr 2026 – Jul 2026

Spaceborn-India (Remote)

- Contributed to end-to-end robotics project execution — from drone frame selection and component integration to physical assembly, testing, and pre-deployment validation.
- Configured, tuned, and tested the PX4 flight-control stack, performing parameter tuning, flight testing, and troubleshooting via QGroundControl in simulated environments.
- Built robotics simulation and validation workflows (ROS2, Gazebo, RViz) to verify system performance before deployment.
- Integrated LiDAR sensors for 3D point cloud generation and environmental perception as part of sensor system commissioning.

- Logged, analysed, and documented flight telemetry data for performance evaluation and project reporting; maintained version control using GitHub.

PROJECTS

6-DOF Robotic Arm — Hardware Design, Build & Motor Control

Arduino Uno | SolidWorks | BCA8016 Motor Drivers | XL4016 Buck Converter | SG90/MG996R Servos | ROS2 | MoveIt

- Designed, built, and commissioned a 6-DOF robotic arm from scratch on Arduino Uno, including SolidWorks modelling, multi-joint servo actuation (SG90/MG996R), and MoveIt-based motion planning for arm control.
- Integrated BCA8016 motor drivers and XL4016 buck converters for stable actuator control; validated performance through hands-on hardware bring-up and debugging.

2-Wheel Self-Balancing Robot — Control System Design (Ongoing)

Arduino/ESP32 | MPU6050 IMU | PID Control | H-Bridge Motor Driver | Embedded C/C++

- Designing a 2-wheel self-balancing robot on an ESP32-S3 with MPU6050 IMU sensor fusion (complementary/Kalman filtering) for real-time tilt-angle estimation, plus an ultrasonic sensor for obstacle avoidance while moving.
- Developing and tuning a PID control loop to drive H-bridge motor outputs for dynamic balance, powered by a custom 18650-cell 3S battery management power supply system.

Power Electronics PCB Design — Boost & Buck Converters

Altium Designer | Cadence | Power Systems | Component Selection & Integration

- Designed and validated boost & buck converter schematics and PCB layouts (Altium Designer, Cadence), selecting components for efficiency, ripple, and load regulation.
- Tested and commissioned prototype boards using an oscilloscope, multimeter, and bench power supply; produced manufacturing-ready Gerber/BOM documentation.

Boost Converter - https://github.com/AhamedRaafiq/Power_Supply_With_Boost_Converter

Buck Converter - https://github.com/AhamedRaafiq/Power_Supply_With_Buck_Converter

GlucoSense — Embedded Sensor System for Non-Invasive Health Monitoring (Final Year Project)

ESP32-S3 | MAX30102 Sensor | Embedded C/Python | Digital Signal Processing | Machine Learning | FastAPI | React

- Built and commissioned a dual-channel embedded sensor acquisition system (ESP32-S3 + MAX30102), integrating hardware, signal processing, and application layers end-to-end.
- Developed a signal-processing pipeline (noise removal, low/high-pass filtering, ensemble averaging, feature extraction) feeding an XGBoost ML model, delivered as a full-stack embedded-to-cloud system (FastAPI, React) with real-time monitoring and validated performance (91.7% clinical accuracy).

GitHub Link - https://github.com/AhamedRaafiq/GlucoSense_FYP

Real-Time 3D IMU Motion Visualizer — Flight Telemetry & Sensor Fusion

ESP32-S3 | ESP-IDF | ROS2 Jazzy | RViz2 | Madgwick AHRS | TF2 | Python/C/C++

- Developed high-speed embedded firmware (ESP-IDF) to stream 100 Hz IMU telemetry over USB-CDC, implementing automated sensor offset calibration and a 4th-order Butterworth low-pass filter for noise reduction.
- Built a ROS2 sensor-fusion pipeline (Madgwick AHRS) with dynamic TF2 broadcasting and a textured URDF model for real-time 6-DOF visualization in RViz2, including rosbag2-based telemetry recording for offline replay.

GitHub Link - https://github.com/AhamedRaafiq/imu_visualizer

Autonomous Drone Simulation, Control & Telemetry Analysis Pipeline

ROS2 | Gazebo | PX4 | QGroundControl | LiDAR | Point Cloud | Telemetry

- Integrated URDF drone airframe models into ROS2 and built a Gazebo simulation for PX4 flight control via QGroundControl, including flight parameter tuning and testing.
- Developed a ROS2 control framework (Python) with custom services, actions, and tf2 transforms, and integrated LiDAR for 3D point cloud environmental perception.

CERTIFICATIONS

- Robotics and ROS2 — Learn by Doing! Manipulators — Udemy
- PCB Design in Cadence OrCAD/Allegro: Basics to Expert Level — Udemy
- PCB Design with Altium Designer — Udemy
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI (Coursera), Jun 2025
- Programming for Everybody (Getting Started with Python) — University of Michigan (Coursera), Aug 2023

LANGUAGES

Tamil (Native) | English (Fluent) | Sinhala (Fluent) | Urdu (Intermediate) | Hindi (Beginner)